

Submission Summary

Conference Name

8th International Conference on Communication and Intelligent Systems

Paper ID

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Paper Title

Digital Forensics in Cybersecurity: Techniques for Effective Evidence Investigation

Abstract

The digital forensics process is vital in today's cybersecurity landscape since it facilitates the identification, collection, preservation, and analysis of digital evidence in cybercrime cases. In the face of rising digital technologies and threats like malware attacks, data breaches, and access violations, it is imperative to have efficient forensic investigation techniques. This research analyzes various methodologies and tools used for advanced digital forensics and their impact on improving digital investigation. An experiment involving simulated cyberattacks was conducted within the framework of virtual machines. The tools like Autopsy and Wireshark were used in collecting and analyzing evidence in the experimental setting. Furthermore, techniques of automation using scripting languages like Python and Bash scripts were employed to speed up the investigation process. The findings show that a well-structured process of digital forensics coupled with automation improves the efficiency of the evidence collection and analysis.

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Authors

Anushka . (Chitkara University) <anushka1322.be22@chitkara.edu.in>

Drishti Bhardwaj (Chitkara University) <drishti306.be22@chitkara.edu.in>

Geetanshu Mantro (Chitkara University) <geetanshu324.be22@chitkara.edu.in>

Vriti Jaitley (Chitkara University) <vriti4862.be22@chitkara.edu.in>

Dr. Shikha Tuteja (Chitkara University) <shikha.1290@chitkara.edu.in>

Primary Subject Area

Intelligent Data Analytics and Computing

Secondary Subject Areas

Cyber Security

Submission Files

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The author confirms sole responsibility for the following: study conception and design, data collection, analysis and interpretation of results, and manuscript preparation.

Study conception and design: Anushka;

Implementation and experiments: Anushka, Vriti Jaitley;

Visualization and documentation: Drishti Bhardwaj, Vriti Jaitley;

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